



Fake Internship & Job Scam Detection System

PART 1 — PROJECT FOUNDATION, SYSTEM DESIGN & DATABASE ARCHITECTURE

1. Project Overview

Project Title: Fake Internship & Job Scam Detection System

Project Category

- Data Analytics
- Data Science
- Cybersecurity
- Fraud Detection
- NLP & Machine Learning
- SQL Database Management

Project Objective

The objective of this project is to build an AI-powered platform capable of detecting fake internship and job postings using machine learning, NLP techniques, SQL databases, web scraping, and fraud risk analysis.

The system will help students and job seekers identify suspicious opportunities before applying.

2. Problem Statement

Thousands of students become victims of fake internship and job scams every year. Fraudsters create fake opportunities to:

- Collect personal information
- Demand registration or training fees



- Conduct phishing attacks
- Steal resumes and identity data
- Manipulate students through unrealistic offers

Most students are unable to differentiate between genuine and fraudulent job opportunities.

This project aims to solve this problem by creating an automated scam detection platform.

3. Proposed Solution

The system will:

1. Scrape internship and job postings from multiple platforms
2. Analyze recruiter and company information
3. Detect suspicious keywords and fraud patterns
4. Check domain reputation and domain age
5. Detect unrealistic salary offers
6. Generate a scam probability score
7. Allow users to report suspicious job postings
8. Maintain a recruiter blacklist database

4. Major Features

4.1 Job Posting Scraper

The system will automatically collect job and internship listings from public websites.

Supported Sources

- LinkedIn Jobs
- Internshala
- Indeed
- Naukri
- Company career pages
- Startup websites



Extracted Data

- Job title
- Company name
- Salary
- Location
- Job description
- Recruiter email
- Company website
- Posting date

4.2 Scam Probability Score

Each job posting will receive a fraud probability score.

Risk Levels

Score Range	Risk Level
0 – 30	Low Risk
31 – 60	Medium Risk
61 – 100	High Risk

4.3 Suspicious Domain Detection

The system will analyze domains using:

- Domain age
- SSL certificate status
- WHOIS information
- Blacklist databases
- Suspicious naming patterns

Example

Suspicious domains:

- careers-google-job.xyz
- amazon-job-hiring-free.net



4.4 Salary Anomaly Detection

The platform will identify unrealistic salary offers.

Example

- ₹80,000/month internship for freshers
- No skills required
- Work only 1 hour/day

Such postings will be flagged as suspicious.

4.5 Blacklisted Recruiter Database

The system will maintain a database of recruiters and companies previously reported for scams.

4.6 Scam Reporting System

Users can report suspicious internships and jobs.

User Submission Fields

- Company name
- Job URL
- Scam description
- Screenshot URL
- Contact details

5. System Architecture

Frontend (HTML/CSS/JavaScript)



Backend API (FastAPI / Flask)



PostgreSQL Database (Supabase)



Machine Learning Fraud Engine



Analytics Dashboard



6. Technology Stack

Component	Technology
Frontend	HTML, CSS, JavaScript
Backend	FastAPI / Flask
Database	PostgreSQL / Supabase
Web Scraping	BeautifulSoup, Selenium, Scrapy
Machine Learning	Scikit-learn
NLP	spaCy, NLTK
Dashboard	Chart.js / Streamlit
Deployment	Render
Version Control	GitHub

7. Frontend Module

Pages

Home Page

Features:

- Search box
- Scam checker
- Statistics section
- Recent alerts

Scam Analysis Page

Displays:

- Scam score
- Domain reputation



- Suspicious keywords
- Salary anomaly status

Recruiter Verification Page

Users can verify:

- Recruiter email
- Company domain
- Previous reports

Scam Report Form

Allows public reporting of fake opportunities.

Admin Dashboard

Displays:

- Total reports
- High-risk domains
- Scam trends
- Recruiter statistics

8. Backend Module

The backend will:

- Receive frontend requests
- Process job analysis
- Execute ML predictions
- Access SQL database
- Return fraud detection results

Major APIs

API	Purpose
/analyze	Analyze job posting
/report	Submit scam report



/recruiter-check Verify recruiter

/dashboard Analytics data

/domain-check Domain analysis

9. Database Design

9.1 job_posts Table

Column	Type
job_id	UUID
title	TEXT
company_name	TEXT
salary	NUMERIC
location	TEXT
description	TEXT
posted_date	DATE
source_url	TEXT
domain_name	TEXT
scam_score	FLOAT
is_flagged	BOOLEAN



9.2 recruiter_profiles Table

Column	Type
recruiter_id	UUID
recruiter_name	TEXT
email	TEXT
company	TEXT
linkedin_url	TEXT
domain_name	TEXT
verified	BOOLEAN
previous_reports	INT

9.3 scam_reports Table

Column	Type
report_id	UUID
job_id	UUID
report_reason	TEXT
user_comment	TEXT
severity	INT
reported_at	TIMESTAMP



9.4 domain_reputation Table

Column	Type
domain_id	UUID
domain_name	TEXT
domain_age_days	INT
ssl_valid	BOOLEAN
trust_score	FLOAT
blacklisted	BOOLEAN

9.5 flagged_keywords Table

Column	Type
keyword_id	UUID
keyword	TEXT
fraud_weight	FLOAT

10. SQL Queries

Top Suspicious Companies

```
SELECT company_name, AVG(scam_score)
FROM job_posts
GROUP BY company_name
ORDER BY AVG(scam_score) DESC;
```



High Risk Domains

```
SELECT domain_name, trust_score  
FROM domain_reputation  
WHERE trust_score < 0.3;
```

Most Common Scam Keywords

```
SELECT keyword, COUNT(*)  
FROM flagged_keywords  
GROUP BY keyword  
ORDER BY COUNT(*) DESC;
```

11. Machine Learning Module

Objectives

The ML system will:

- Detect fraudulent job descriptions
- Identify suspicious patterns
- Predict scam probability

11.1 NLP Fraud Detection

Techniques

- TF-IDF
- Tokenization
- Keyword extraction
- Text classification

Libraries

- Scikit-learn
- spaCy
- NLTK

11.2 Fraud Classification Models

Suggested Models

- Logistic Regression
- Random Forest
- XGBoost
- Naive Bayes



11.3 Salary Anomaly Detection Algorithms

- Isolation Forest
- Z-score detection

11.4 Risk Scoring Formula

Risk Score =

(Keyword Fraud Score $\times$ 0.4)
+ (Domain Risk $\times$ 0.3)
+ (Salary Anomaly $\times$ 0.2)
+ (User Reports $\times$ 0.1)

12. Web Scraping Module

Tools

- BeautifulSoup
- Selenium
- Scrapy

Extracted Information

- Job descriptions
- Recruiter details
- Company websites
- Salary information
- Contact details

13. Deployment Strategy

Database Hosting

Supabase PostgreSQL

Deployment Components

Service

Frontend

Backend API

Platform

Render Static Site

Render Web Service



Database

Supabase

Scheduled Scraping

GitHub Actions / Cron Jobs

14. Security Features

- Input validation
- SQL injection protection
- API rate limiting
- Secure password hashing
- HTTPS support
- Spam prevention

15. Dashboard Analytics

Dashboard Metrics

- Total analyzed jobs
- Total scam reports
- Scam percentage
- Top scam domains
- High-risk recruiters
- Daily fraud trends

Visualization Tools

- Chart.js
- Streamlit

16. Expected Outcomes

The project will:

- Help students identify fake opportunities
- Reduce internship fraud
- Build awareness about scam patterns
- Create a centralized fraud reporting system
- Provide hands-on learning for interns



17. Future Enhancements

Browser Extension

Display scam score directly on job websites.

AI Chatbot

Allow users to ask:

"Is this internship safe?"

Email Scam Detection

Analyze recruiter emails automatically.

Mobile Application

Android/iOS app for quick verification.

Real-Time Notifications

Alert users when suspicious jobs are detected.

18. Advantages of the Project

- Real-world problem solving
- Strong portfolio project
- Multi-domain learning
- AI + SQL + Analytics integration
- Scalable architecture
- Industry-relevant skills



PART 2 — AI MODULES, DEPLOYMENT, SECURITY & ADVANCED ANALYTICS

19. Project Workflow

Complete Operational Workflow

Step 1 — Data Collection

The system continuously collects internship and job postings from multiple online platforms using automated scraping modules.

Data Sources

- Internship portals
- Job portals
- Startup websites
- Company career pages
- Public recruitment listings
- Social media recruitment posts

Collected Attributes

- Job title
- Company information
- Recruiter details
- Salary package
- Skills required
- Application link
- Posting date
- Email address
- Domain name
- Location



Step 2 — Data Cleaning & Preprocessing

Raw scraped data often contains:

- Duplicate records
- HTML tags
- Broken URLs
- Missing values
- Spam content
- Invalid recruiter information

Preprocessing Operations

- Text normalization
- Stopword removal
- Lowercasing
- URL validation
- Duplicate filtering
- Null handling
- Tokenization
- Regex cleaning

Step 3 — Database Storage

After preprocessing, the cleaned data is stored inside PostgreSQL tables.

Storage Pipeline

Scraper → Data Cleaning → Validation → PostgreSQL Database

Step 4 — Feature Engineering

The ML system extracts fraud-related features from job postings.

Engineered Features

Feature	Description
Keyword Risk Score	Fraud-related keywords
SalaryRisk Score	Unrealistic salary offers



Domain Trust Score	Domain reputation
Recruiter Trust Score	Recruiter authenticity
Grammar Quality Score	NLP-based language analysis
Domain Age	Newly registered domains
Contact Authenticity	Personal email detection
Posting Frequency	Spam posting behavior

Step 5 — Fraud Prediction

Machine learning models analyze all extracted features and generate a scam probability score.

Prediction Categories

- Safe
- Suspicious
- High Risk
- Confirmed Scam

Step 6 — Dashboard Analytics

The analytics dashboard displays:

- Fraud trends
- Scam distribution
- High-risk domains
- Recruiter risk analytics
- Daily scraping statistics
- Model prediction accuracy

20. Detailed Frontend Documentation

Frontend Objective

The frontend provides a user-friendly interface for:

- Scam detection
- Recruiter verification



- Fraud reporting
- Dashboard analytics

21.1 Home Page

Components

Navigation Bar

Contains:

- Home
- Scam Checker
- Verify Recruiter
- Report Scam
- Dashboard

Hero Section

Displays:

- Main headline
- Scam awareness information
- CTA buttons

Live Statistics

Displays:

- Total analyzed jobs
- Fake jobs detected
- Verified recruiters
- Total scam reports

Recent Scam Alerts

Shows latest detected scam postings.

21.2 Scam Checker Page

User Inputs

- Job URL
- Job description
- Recruiter email



Analysis Results

- Scam score
- Risk category
- Suspicious keywords
- Domain trust score
- Salary anomaly status
- Recommendation

21.3 Recruiter Verification Page

Verification Checks

- Domain verification
- Email authenticity
- Company matching
- Previous reports
-

Result Types

- Verified
- Suspicious
- Blacklisted

21.4 Public Reporting Form

Fields

- Company name
- Job title
- Job URL
- Scam type
- Screenshot upload URL
- Contact information
- Description



21.5 Admin Dashboard

Dashboard Sections

Fraud Analytics

- Total scam posts
- Monthly fraud growth
- Top risky companies

Domain Monitoring

- Suspicious domains
- Blacklisted domains
- Newly detected domains

Recruiter Monitoring

- Recruiter activity
- Spam frequency
- Fraud reports

ML Analytics

- Prediction accuracy
- False positives
- Model confidence

22. Detailed Backend Documentation

Backend Responsibilities

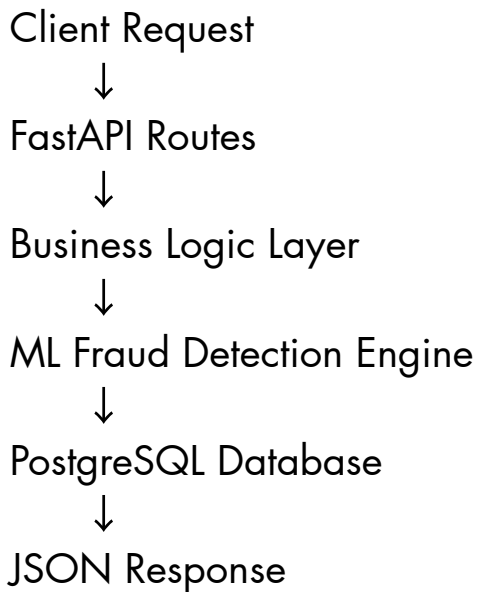
The backend acts as the central processing layer.

Functions

- API management
- ML prediction execution
- Database operations
- Authentication
- Validation
- Data aggregation



Backend Architecture



Authentication System

User Roles

Role	Permissions
User	Scam checking
Reporter	Submit reports
Admin	Full access
Analyst	Dashboard analytics

API Response Example

```
{  
  "job_title": "Data Entry Internship",  
  "company": "XYZ Solutions",  
  "scam_score": 82,  
  "risk_level": "HIGH",  
  "domain_age": 15,
```



```
"suspicious_keywords": [
  "registration fee",
  "instant joining"
]
```

23. Database Relationships

ER Relationship Overview



Foreign Key Relationships

Parent Table	Child Table	Relation
job_posts	scam_reports	One-to-Many
recruiter_profiles	job_posts	One-to-Many
domain_reputation	recruiter_profiles	One-to-Many

24. NLP System Documentation

NLP Objective

The NLP module identifies fraudulent language patterns.

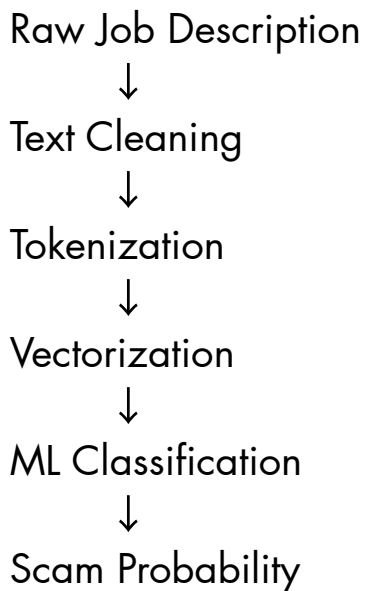
Common Fraud Keywords

Keyword	Fraud Weight
registration fee	0.95



instant joining	0.82
earn daily	0.77
limited seats	0.69
WhatsApp HR	0.91
no interview	0.75

NLP Pipeline



NLP Feature Extraction

Techniques

- TF-IDF
- Bag of Words
- N-grams
- Named Entity Recognition
- Sentiment Analysis



25. Machine Learning Training Documentation

Dataset Requirements

Genuine Job Data

- Verified companies
- Official internship portals
- Government job listings

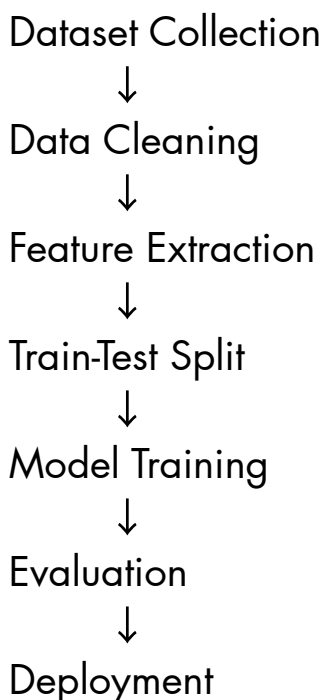
Scam Data

- User reports
- Scam complaints
- Fraud job screenshots
- Blacklisted domains

Data Labeling

Label	Meaning
0	Safe
1	Scam

Training Workflow





Evaluation Metrics

Metric	Purpose
Accuracy	Overall performance
Precision	Correct scam predictions
Recall	Scam detection capability
F1 Score	Balanced evaluation

26. Dashboard Analytics Documentation

Visualization Components

Pie Charts

- Scam vs genuine jobs

Line Charts

- Monthly scam trends

Heatmaps

- Scam density by region

Bar Charts

- Most common scam keywords

Tables

- High-risk recruiters
- Blacklisted domains

27. Security & Compliance Documentation

Security Risks

Potential Threats

- SQL injection
- Spam submissions



- API abuse
- Bot attacks
- Data scraping abuse

Protection Mechanisms

Backend Security

- Prepared SQL statements
- Authentication middleware
- Input validation
- Rate limiting
- JWT authentication

Frontend Security

- Form validation
- CAPTCHA integration
- CSRF protection

28. Performance Optimization

Database Optimization

Techniques

- Indexing
- Query optimization
- Connection pooling
- Caching

Backend Optimization

Techniques

- Async APIs
- Background tasks
- Batch processing
- Pagination



Frontend Optimization

Techniques

- Lazy loading
- Minified assets
- Image optimization
- Browser caching

29. Testing Documentation

Testing Types

Testing	Purpose
Unit Testing	Individual functions
API Testing	Endpoint validation
UI Testing	Frontend functionality
Database Testing	Query verification
ML Testing	Model accuracy

Example Test Cases

Scam Detection Test

Input	Expected Result
Registration fee internship	High Risk
Official company internship	Safe

30. Limitations of the System

- Scraping restrictions on some platforms
- False positives possible
- Limited access to private recruiter data
- New scam patterns may bypass models initially



31. Scalability Planning

Future Scaling Strategy

Infrastructure Scaling

- Docker containers
- Kubernetes deployment
- CDN integration
- Distributed scraping system

AI Scaling

- Transformer models
- Deep learning pipelines
- Real-time fraud detection

32. Business & Social Impact

Educational Impact

- **Student safety awareness**
- **Scam prevention education**
- **Career platform trust improvement**

Industry Impact

- Safer recruitment ecosystem
- Fraud monitoring automation
- Data-driven scam analytics

33. Resume & Portfolio Benefits

Interns working on this project gain experience in:

- SQL database design
- Web scraping
- Machine learning
- NLP systems
- Fraud detection
- Dashboard analytics
- API development
- Full-stack development
- Production deployment



34. Conclusion

The Fake Internship & Job Scam Detection System is a comprehensive AI-powered fraud detection platform developed to protect students and job seekers from fake opportunities.

The project combines:

- Artificial Intelligence
- SQL databases
- NLP systems
- Fraud analytics
- Full-stack web development
- Dashboard visualization
- Real-world cybersecurity concepts

The system is scalable, industry-relevant, and highly valuable for internship training, research, analytics learning, and production-level development.

35. References

Technologies

- Python
- FastAPI
- PostgreSQL
- Supabase
- Scikit-learn
- spaCy
- BeautifulSoup
- Selenium
- Render
- Chart.js
- GitHub



Learning Resources

- [Official Python Documentation](#)
- [PostgreSQL Documentation](#)
- [FastAPI Documentation](#)
- [Scikit-learn Documentation](#)
- [spaCy Documentation](#)
- [Supabase Documentation](#)
- [Render Documentation](#)